Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	5 April 2026
Team ID	SWTID-2026-8642
Project Name	Streamlining Ticket Assignment for Efficient Support Operations.
Maximum Marks	4 Marks

Technical Architecture:

The technical architecture for the project **"Streamlining Ticket Assignment for Efficient Support Operations"** is designed to ensure efficiency, scalability, and reliability in handling support requests. The system follows a layered architecture consisting of a user interface layer, application layer, and data layer. The user interface allows customers and support agents to create, view, and manage tickets through web or mobile applications. The application layer processes ticket data, applies business rules, and automatically assigns tickets to appropriate agents based on predefined criteria such as priority, category, and workload. The backend is supported by a centralized database that stores user details, ticket information, and activity logs. Integration with notification services enables real-time alerts via email or SMS, ensuring timely responses. This architecture ensures smooth communication, faster resolution times, and improved overall support efficiency.

The guidelines for the project *Streamlining Ticket Assignment for Efficient Support Operations* ensure smooth and efficient functioning of the system. Users should register with valid details and securely log in to access the system. While creating tickets, users must provide clear descriptions, select appropriate categories, and set priority levels. Tickets should be assigned automatically based on workload or manually by the admin to ensure fair distribution. Users must be able to track ticket status, and support agents should update progress regularly until resolution. The system must follow strong security practices, maintain fast performance, and ensure 24/7 availability. Regular maintenance and monitoring are required to keep the system reliable and efficient.

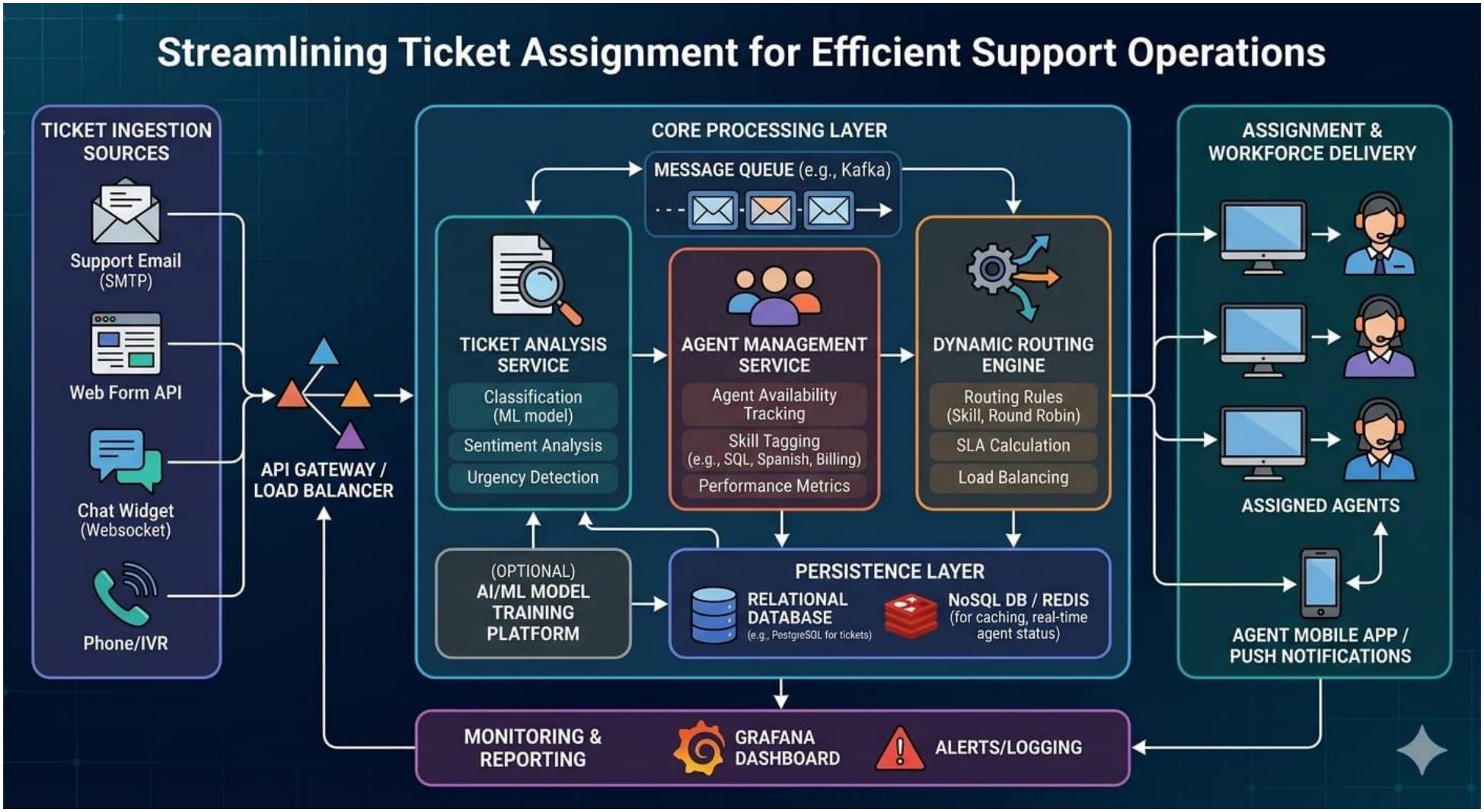
Streamlining Ticket Assignment for Efficient Support Operations
TICKET INGESTION SOURCES
Support Email (SMTP)
Web Form API
Chat Widget (Websocket)
Phone/IVR
API GATEWAY / LOAD BALANCER
CORE PROCESSING LAYER
MESSAGE QUEUE (e.g., Kafka)
TICKET ANALYSIS SERVICE
Classification (ML model)
Sentiment Analysis
Urgency Detection
AGENT MANAGEMENT SERVICE
Agent Availability Tracking
Skill Tagging (e.g., SQL, Spanish, Billing)
Performance Metrics
DYNAMIC ROUTING ENGINE
Routing Rules (Skill, Round Robin)
SLA Calculation
Load Balancing
(OPTIONAL) AI/ML MODEL TRAINING PLATFORM
PERSISTENCE LAYER
RELATIONAL DATABASE (e.g., PestgreSQL for tickets)
NoSQL DB / REDIS (for caching, real-time agent status)
ASSIGNMENT & WORKFORCE DELIVERY
ASSIGNED AGENTS
AGENT MOBILE APP / PUSH NOTIFICATIONS
MONITORING & REPORTING
GRAFANA DASHBOARD
ALERTS/LOGGING

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interface for users (customers, agents, admin) to interact with the system (Web App / Dashboard)	HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic-1	Handles user registration, login, and authentication processes	Java / Python
3.	Application Logic-2	Ticket creation, categorization, and priority assignment logic	Python / Node.js
4.	Application Logic-3	Automated ticket assignment using rules or AI-based decision making	Machine Learning / Rule-based Engine
5.	Database	Stores user data, ticket details, status, and logs	MySQL, NoSQL, etc.
6.	Cloud Database	Cloud-based storage for scalability and remote access	Firebase / AWS RDS / MongoDB Atlas
7.	File Storage	Stores attachments such as screenshots, documents in tickets	AWS S3 / Local Storage
8.	External API-1	Email and notification service for alerts and updates	SMTP / SendGrid API.
9.	External API-2	Integration for authentication or third-party login (Gmail, LinkedIn)	Google API / LinkedIn API
10.	Machine Learning Model	Predicts ticket priority and assigns to appropriate support agents	Classification Model (AI/ML)
11.	Infrastructure (Server / Cloud)	Deployment and hosting of application	AWS / Azure / Google Cloud / Local Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used for building the web-based ticket management system	React JS (Frontend), Node.js / Django (Backend)

S.No	Characteristics	Description	Technology
2.	Security Implementations	Secure authentication, data protection, and access control for users and admin	JWT Authentication, SHA-256 Encryption, HTTPS, Role-Based Access Control (RBAC)
3.	Scalable Architecture	System designed to handle increasing users and tickets efficiently using modular architecture	3-Tier Architecture (UI, Logic, Database), Microservices (optional), Docker
4.	Availability	Ensures system is always accessible with minimal downtime	Cloud Hosting (AWS/Azure), Load Balancer, Distributed Servers
5.	Performance	Optimized for fast response and handling multiple requests simultaneously	Caching (Redis), CDN, Optimized Database Queries